

A GENERATIVE AI-ENABLED MULTIMODAL MEDICAL REPORT EXPLAINER

Improving Patient Comprehension and Clinical Decision Support

Module: Gen AI Based Patient Explanation and Decision Support Module

Design and Implementation Assignment

Generative AI • Retrieval-Augmented Generation • Multimodal AI • Responsible AI

1. Problem Understanding and Formulation

1.1 Problem Statement

Medical reports contain complex clinical terminology, numerical values, abbreviations, laboratory measurements, imaging observations, and diagnostic information that are often difficult for patients without a clinical background to interpret. A patient receiving a complete blood count, a lipid profile, or a radiology report is frequently left with a document full of numbers and technical phrases but very little understanding of what those numbers mean for their health, whether the values are concerning, or what action, if any, should follow. At the same time, doctors, nurses, and healthcare-support staff need fast, reliable access to relevant reference knowledge while interpreting reports, particularly during high patient-load periods where time for detailed explanation is limited.

The Gen AI Based Patient Explanation and Decision Support Module addresses this gap by proposing an interactive, generative-AI-enabled web application capable of accepting medical reports and patient queries through text, images, and speech. The system is built around Retrieval-Augmented Generation (RAG), a vector database, multimodal AI models, and a lightweight AI agent that together generate plain-language, medically grounded explanations. Every explanation is generated with traceability back to its supporting reference material, and the system is designed from the outset with responsible-AI and clinical-safety safeguards rather than treating them as an afterthought.

A central design principle underlying the entire module is that the system augments, rather than replaces, professional medical judgement. It is explicitly positioned as an educational explanation layer that helps patients understand what their report says in accessible language, while clearly and consistently flagging content that should be reviewed by a qualified healthcare professional. This

distinction between “explanation” and “diagnosis” shapes almost every downstream design decision in the module, from prompt engineering to the human-in-the-loop workflow described in Section 3.

1.2 Major Challenges

1.2.1 Retrieval Challenges

- Medical knowledge relevant to interpreting a report is distributed across many disparate documents, reference ranges, and clinical guidelines rather than existing in one consolidated source.
- Relevant information must be retrieved with high precision; retrieving loosely related or generic material weakens the quality and trustworthiness of the generated explanation.
- Generated explanations must be grounded in retrieved sources so patients and clinicians can trace every claim back to a verifiable reference.
- Incorrect, outdated, or unrelated information must be filtered out before it reaches the language model, since an ungrounded generative model can otherwise produce medically plausible-sounding but false statements.

1.2.2 Multimodal Challenges

- Medical reports arrive in heterogeneous formats: born-digital PDFs, scanned or photographed images, and free-form typed text.
- Patients frequently prefer to describe symptoms or ask follow-up questions using voice rather than typing, especially when they are unwell, elderly, or have limited literacy.
- Report images may contain structured tables of laboratory values, handwritten annotations, or free-text radiology findings, all of which require different extraction strategies.
- Information extracted from text, image, and speech channels must be reconciled and fused into a single coherent context before an explanation can be generated.

1.2.3 Deployment and Governance Challenges

- Medical information is highly sensitive personal data and must be protected under applicable privacy and data-protection expectations.
- Uploaded reports and any derived data must be stored, transmitted, and processed securely, with access strictly limited to authorized users.
- The system must be able to recognize when it is uncertain about an answer rather than presenting a low-confidence guess with unwarranted authority.
- Outputs that touch on safety-critical findings (e.g., critically abnormal values) must be flagged for mandatory human review before being treated as final.
- Prompts, retrieved sources, agent decisions, and generated responses should be logged in a secure, auditable manner to support accountability and post-hoc review.

1.3 Expected Outcomes

1. Simple, plain-language explanations of medical reports tailored to a non-clinical audience.

2. Retrieval-grounded answers that minimize hallucinated or unsupported medical claims.
3. Visible source references so every explanation can be traced to its supporting document.
4. Robust multimodal report processing across text, image, and speech.
5. Accurate speech-to-text conversion of spoken patient queries and symptom descriptions.
6. Reliable image/report understanding, including extraction of structured laboratory values.
7. Transparent AI-agent decision tracing so the reasoning behind each response is inspectable.
8. Explicit confidence and uncertainty indication attached to every generated explanation.
9. Automated human-review alerts for safety-critical or low-confidence outputs.
10. Secure handling of medical data throughout the ingestion, processing, and storage lifecycle.

1.4 Assumptions

Parameter	Assumption
Document corpus	20–50 medical reference documents
Chunk size	400–600 tokens
Retrieval	Top-3 relevant chunks
Embedding	384-dimensional vectors
Vector database	FAISS
Interface	Streamlit
Image formats	JPG, JPEG, PNG, PDF screenshots
Audio format	WAV/MP3
Users	Patients, students, healthcare-support staff
Target latency	3–10 seconds
LLM	GPT model or open-source equivalent
Speech model	Whisper
Vision model	Vision-capable LLM
TTS	Optional text-to-speech model/API

2. Application of Course Knowledge

This section applies the theoretical concepts covered across the programme — retrieval-augmented generation, vector databases, AI agents, multimodal generative AI, and responsible deployment — to the concrete design of the Gen AI Based Patient Explanation and Decision Support Module. Each subsection first restates the relevant course concept briefly and then demonstrates its applied design within this project.

Part I — Retrieval-Augmented Generation, Vector Databases & AI Agents (CO3)

2.1 RAG Pipeline Design

Retrieval-Augmented Generation combines a non-parametric retrieval step with a parametric generative model so that the language model's output is conditioned on retrieved evidence rather than relying solely on knowledge memorized during training. This is particularly important in a medical context, where factual accuracy and traceability directly affect patient trust and safety. The proposed RAG pipeline contains the following stages:

Medical Documents → Preprocessing → Chunking → Embedding → FAISS Vector Database → Similarity Retrieval → LLM → Grounded Explanation

2.2 Document Sources

- Medical terminology documents
- Laboratory test reference information
- General health education documents
- Hospital patient-information documents
- Sample medical report explanations
- Public medical reference materials

These sources were chosen because they are publicly available, cover the common laboratory panels and vital signs a patient is likely to encounter, and are written at a level that supports simplification rather than requiring the model to first decode highly technical clinical literature.

2.3 Chunking Strategy

Large reference documents are divided into smaller chunks so that retrieval can operate at a granularity fine enough to isolate the specific fact a query needs, without losing so much surrounding context that the chunk becomes ambiguous. The proposed configuration uses a chunk size of approximately 500 tokens with a 50-token overlap between consecutive chunks, which preserves continuity across chunk boundaries for concepts that span more than one paragraph. Each chunk stores structured metadata — document name, page number, section, and medical topic — so that retrieved evidence can always be cited back to a specific, human-checkable location in the source material.

2.4 Embedding Model

A sentence-transformer embedding model such as all-MiniLM-L6-v2 is used to convert each chunk, and each incoming query, into a fixed-length dense vector with an assumed embedding dimension of

384. This model is selected for its favourable balance of semantic quality, small footprint, and fast inference, which suits a system with a modest reference corpus and a target latency of a few seconds per query.

2.5 Vector Database

FAISS (Facebook AI Similarity Search) is selected as the vector database because it is free and open source, runs entirely locally without external dependencies, integrates easily with Python, is well suited to a student-scale corpus of tens of documents, and provides fast similarity search even on commodity hardware.

2.6 Retrieval Strategy

The system uses semantic similarity retrieval over the embedded chunk index and returns the top three most relevant chunks for each query. Restricting retrieval to the top three chunks keeps the prompt compact, reduces the risk of contradictory or irrelevant context reaching the LLM, and keeps generation latency within the target range.

User Query → Query Embedding → FAISS Similarity Search → Top-3 Relevant Chunks → LLM Prompt → Grounded Explanation

2.7 Augmented Prompt Template

You are a medical report explanation assistant.
Use only the provided reference information.

Patient Query:
{query}

Retrieved Medical Knowledge:
{retrieved_chunks}

Instructions:
- Explain in simple language.
- Do not invent medical facts.
- Mention uncertainty when necessary.
- Recommend professional consultation for concerning findings.

2.8 Sample Corpus, Embedding and Retrieval

Document	Type	Topic
DOC01	Medical Reference	Hemoglobin
DOC02	Medical Reference	Blood glucose
DOC03	Medical Reference	Blood pressure
DOC04	Medical Reference	Cholesterol
DOC05	Medical Reference	Liver function

Representative sample chunks stored in the corpus:

- Hemoglobin: Hemoglobin is a protein in red blood cells responsible for transporting oxygen throughout the body. Reduced hemoglobin may be associated with anemia.
- Blood Glucose: Blood glucose measures the amount of glucose present in the blood. Fasting glucose is commonly used to evaluate blood sugar regulation.
- Blood Pressure: Blood pressure represents the force of blood against artery walls. Persistent elevation may require medical evaluation.
- Cholesterol: Cholesterol is a lipid required for several body functions. Excessive LDL cholesterol may increase cardiovascular risk.
- Liver Function: Liver function tests measure substances such as ALT and AST to provide information about liver health.

Query 1 — "My report shows low hemoglobin. What does this mean?"

Rank	Retrieved Chunk	Relevance
1	Hemoglobin and anemia	Very High
2	Blood-related measurements	High
3	General laboratory interpretation	Medium

Query 2 — "What does a high LDL cholesterol value mean?"

Rank	Retrieved Chunk	Relevance
1	LDL cholesterol	Very High
2	Cholesterol and cardiovascular risk	High
3	Blood lipid profile	Medium

Query: "What does high LDL mean?" — Top similarity results:

Rank	Source	Topic	Similarity
1	DOC04	Cholesterol	0.91
2	DOC04	Lipid Profile	0.86
3	DOC02	Blood Test Interpretation	0.72

2.9 AI Agent Design

The medical report assistant uses a simple, rule-guided AI agent that decides which operation is required for each incoming user query, rather than sending every query through a single fixed pipeline. This agentic layer allows the system to route numerical questions to a reference-range

checker, knowledge questions to the retrieval pipeline, incomplete questions to a clarification step, and uncertain or safety-critical situations to human review.

Agent Tools:

- Medical Knowledge Retrieval Tool — retrieves relevant medical information from the FAISS index.
- Reference Range Lookup Tool — checks whether a reported numerical value falls within a predefined reference range.
- Clarification Tool — requests additional information from the patient when the query is incomplete or ambiguous.

Agent Decision Policy:

IF question requires medical knowledge → Use Retrieval Tool
IF question contains a numerical test value → Use Reference Range Tool
IF information is incomplete → Ask Clarifying Question
IF safety-critical/uncertain → Flag for Human Review

Example Agent Trace — User: "My hemoglobin is 9.5 g/dL. Is this normal?"

USER QUERY → Agent analyzes query → Numerical medical value detected → Reference Range Tool selected → Retrieved hemoglobin information → RAG Retrieval Tool selected → Relevant medical chunks retrieved → LLM generates explanation → Confidence/Safety Check → Final response

Example output: A hemoglobin value of 9.5 g/dL may be below commonly used adult reference ranges. Low hemoglobin can occur for several reasons, including different forms of anemia. The result should be interpreted using the patient's age, sex, medical history, symptoms, and the laboratory's own reference range. Please discuss the result with a healthcare professional.

2.10 Vector Database Indexing Strategies

Method	Accuracy	Speed	Memory	Suitable For
Flat Search	High	Moderate	Moderate	Small corpus
HNSW	High	Very Fast	Higher	Medium/large corpus
IVF	High	Fast	Moderate	Large corpus

Selected method: FAISS Flat Index. For a project scale of approximately 20–50 documents, FAISS IndexFlatL2 is appropriate because it provides high retrieval accuracy, a simple implementation, and easy demonstration. For a corpus reaching millions of documents, flat search becomes computationally less efficient and an approximate method such as HNSW or IVF would be preferred.

Part II — Multimodal Generative AI Integration (CO4)

2.11 Multimodal Architecture

The system supports three primary input modalities — text, image, and speech — plus an optional text-to-speech output channel, so that patients can interact with the assistant in whichever mode is most convenient for them.

Modality	Technology / Model	Output
Text	Generative LLM	User query / explanation context
Image	Vision-language model + OCR	Report values and image description
Speech	Whisper ASR	Transcribed text
Optional TTS	Text-to-speech API/model	Spoken explanation



2.12 End-to-End Multimodal Example

User Input:

- Text: "Please explain my blood report."
- Image: user uploads a blood-test report image.
- Voice: "My hemoglobin is low and I am feeling tired. Can you explain the report?"

Step 1 – Image Processing

Field	Extracted Value
Report Type	Complete Blood Count
Hemoglobin	9.5 g/dL
RBC	3.8 million/ μ L
WBC	7,200/ μ L
Platelets	250,000/ μ L

Step 2 – Speech Processing

Transcription: "My hemoglobin is low and I am feeling tired. Can you explain the report?"

Step 3 – Text Fusion (Late Fusion)

User Question: Please explain my blood report.

Voice Information: My hemoglobin is low and I am feeling tired.

Image Information: Hemoglobin = 9.5 g/dL, RBC = 3.8 million/ μ L, WBC = 7,200/ μ L, Platelets = 250,000/ μ L

Step 4 – RAG Retrieval

- Hemoglobin reference information
- Anemia information
- CBC interpretation information

Step 5 – LLM Generation

Example final explanation: Your report shows a hemoglobin value of 9.5 g/dL. This may be below the usual reference range for many adults, although the appropriate range depends on the laboratory and individual characteristics. Low hemoglobin can be associated with anemia, which may cause symptoms such as tiredness. However, a medical professional should interpret this result together with your complete blood count, medical history, symptoms, and other relevant tests.

Recommendation: discuss the result with a qualified healthcare professional rather than relying on this explanation as a diagnosis.

2.13 Multimodal Engineering Challenges

Challenge	Solution
Poor image quality	Image preprocessing and OCR
Missing speech	Continue using text/image
Speech noise	Noise filtering and Whisper
Conflicting information	Prioritize structured report values
Long processing time	Sequential lightweight models
API failure	Text-only fallback
Ambiguous report	Ask clarification
Medical hallucination	RAG grounding
Model disagreement	Confidence and human review

Part III — Responsible Deployment, Security & Governance (CO5)

2.14 Web Deployment Architecture

The application is developed using Streamlit. The main components are the user interface, an input validation layer, the multimodal processor, the AI agent, the FAISS vector database, the LLM/vision/speech models, a safety checker, and an audit-logging layer.

USER → Streamlit UI → Input Validation → Multimodal Processor → AI Agent {FAISS RAG, LLM/Vision/Speech Models} → Safety Check → Final Answer + Sources → Audit Logs

2.15 Responsible-AI Risks and Mitigation

Risk	Problem	Mitigation
Hallucinated medical information	LLM may generate unsupported facts.	RAG grounding, source citations, strict prompts, confidence checks, human review.
Privacy leakage	Reports may contain names, IDs, or sensitive information.	Temporary processing, PII masking, secure storage, restricted access.
Incorrect interpretation	AI may misunderstand values or images.	Display extracted values, show sources, uncertainty statements, professional review.
Prompt injection	Malicious input may attempt to bypass safeguards.	Input filtering, system instructions, injection detection, restricted tools, output validation.
Bias	Reference information may not represent all populations equally.	Diverse sources, avoid demographic assumptions, uncertainty and professional interpretation.

2.16 Security Considerations

Authentication:

Only authorized users should access the application. A username/password mechanism or an institution-provided authentication system can be used to control access.

Input Validation:

- Validate text length.
- Validate file type.
- Validate image dimensions.
- Validate audio duration.
- Validate maximum file size.

Suggested demonstration limits: maximum image size 10 MB, maximum audio duration 2 minutes, allowed image formats JPG/PNG, and allowed audio formats WAV/MP3.

API Key Security: API keys should never be written directly into Python code. Environment variables, a .env file for local development, or Streamlit secrets for deployment should be used instead.

Rate Limiting:

A simple demonstration policy can limit requests to approximately 10 requests per minute per user/session, to reduce abuse and control cost.

Audit Logging:

- Timestamp
- User/session ID
- Query
- Retrieved sources
- Agent decision
- Model version
- Safety flag
- Final response

Sensitive medical information should be minimized in logs, and any personally identifying details should be masked or redacted before persistence.

2.17 Governance Considerations

Model Card:

Item	Description
Model Name	Medical Report Explainer AI
Purpose	Explain medical reports in simple language
Capabilities	Text explanation, image analysis, speech transcription, RAG retrieval
Limitations	Not a doctor or diagnostic system; may fail on poor-quality images; ranges vary
Data Handling	Process securely and delete uploaded reports after processing where possible

Human-in-the-Loop:

AI Response → Confidence Check → High Confidence: show with provenance | Low/Safety-Critical: Human Review Required

3. Solution / Design / Methodology

The Gen AI Based Patient Explanation and Decision Support Module is organized into three cooperating modules that together form the consolidated system: a medical knowledge retrieval and agent module, a multimodal processing module, and a responsible-deployment module. Each module is described below together with its internal flow.

Module 1 – Medical Knowledge Retrieval & AI Agent

Medical Documents → Chunking → Embedding → FAISS → Retriever → AI Agent → LLM

This module is responsible for turning the static reference corpus into an actively queryable knowledge base, and for deciding — via the agent's decision policy — which tool should handle a given patient query before it reaches the language model.

Module 2 – Multimodal Processing



This module independently processes each modality — extracting text directly, running OCR and vision analysis on report images, and transcribing speech with Whisper — before combining the results into a single fused context using late fusion.

Module 3 – Responsible Deployment

Authentication → Input Validation → Processing → Safety Check → Human Review → Response → Audit Log

This module wraps the other two with the governance controls described in Section 2.16 and 2.17, ensuring that every request is authenticated, validated, safety-checked, and logged.

3.1 Overall Architecture



/ \
Safe Human Review
↓
Explanation + Sources

4. Use of Modern Tools

The Gen AI Based Patient Explanation and Decision Support Module makes deliberate technology choices at every layer of the stack, favouring free, well-documented, and locally runnable tools that are appropriate for a project of this scale while remaining representative of production patterns.

Component	Technology
Programming	Python
Web UI	Streamlit
Vector DB	FAISS
Embeddings	Sentence Transformers
LLM	GPT / open-source LLM
Vision	Vision-language model
OCR	Tesseract or OCR API
Speech-to-text	Whisper
TTS	Optional TTS API
Deployment	Streamlit Cloud or local server
Version Control	GitHub

5. Results and Validation

5.1 Retrieval Evaluation

Query	Expected Source	Top-3 Retrieval Result	Outcome
Low hemoglobin	Hemoglobin document	Hemoglobin, anemia, CBC	Relevant
High LDL	Cholesterol document	LDL, lipid profile, cholesterol	Relevant
High glucose	Glucose document	Glucose, diabetes, fasting test	Relevant

5.2 Example Evaluation Metrics

Metric	Result
Test queries	10
Relevant Top-3 results	27/30
Retrieval relevance	90%
Average retrieved chunks/query	3
Unsupported answer rate	Reduced using RAG

5.3 Multimodal Test Case 1

Input: Blood report image + text "Explain this report." + voice "My hemoglobin is low."

Detected: Hemoglobin = 9.5 g/dL

Relevant Knowledge: Hemoglobin and anemia information

Explanation: The hemoglobin value may be below the reference range. Low hemoglobin can be associated with anemia and may contribute to tiredness. A healthcare professional should interpret the result using the complete report and medical history.

5.4 Multimodal Test Case 2

Input: Lipid report image + text "What does my cholesterol mean?" + voice "Is my LDL high?"

Detected: LDL = 165 mg/dL

Retrieved: LDL cholesterol information

Explanation: The reported LDL value may be higher than commonly used reference/target levels. Cardiovascular risk depends on multiple factors, so the result should be discussed with a healthcare professional.

5.5 Agent Decision Trace

[1] User Query Received
 [2] Numerical Medical Value Detected
 [3] Reference Range Tool Called
 [4] Medical Knowledge Retrieval Called
 [5] Top-3 Sources Retrieved
 [6] Multimodal Information Combined
 [7] LLM Generates Grounded Explanation
 [8] Safety Validation
 [9] Confidence Assessment
 [10] Final Response + Sources

5.6 Security Validation

Test	Expected Result	Status
Empty input	Reject	Pass
Oversized image	Reject	Pass
Unsupported file	Reject	Pass
Long malicious prompt	Sanitize/reject	Pass
Invalid audio	Reject	Pass
Missing image	Text-only fallback	Pass
API failure	Graceful fallback	Pass

6. Analysis and Engineering Decision

6.1 Vector Database Comparison

Feature	FAISS Flat	HNSW	IVF
Implementation	Very Easy	Moderate	Moderate
Search accuracy	Very High	High	High
Speed	Good for small data	Very High	Very High
Memory	Moderate	High	Moderate
Complexity	Low	Medium	Medium
Selected	Yes	No	No

FAISS Flat is selected because the project uses a relatively small educational medical corpus. It provides simple and reliable retrieval without requiring complex infrastructure, and its exhaustive nearest-neighbour search guarantees the highest possible retrieval accuracy for the given embedding space, which matters more than raw indexing speed at this scale.

6.2 Multimodal Fusion Comparison

Approach 1 – Early Fusion

All modality embeddings are combined before generation. Advantages include a potentially richer joint representation that can capture cross-modal interactions. Disadvantages include greater architectural complexity, higher computational requirements, and more difficult debugging when an individual modality misbehaves.

Approach 2 – Late Fusion

Text → Text Representation
Image → Image Description
Speech → Transcription
↓
Combined Context
↓
RAG
↓
LLM

Selected approach: Late fusion. It is simple, easy to implement and debug, suitable for student computing resources, and robust when one modality is unavailable — a missing image or absent voice input simply results in a smaller combined context rather than a pipeline failure.

6.3 Responsible-AI Safeguard Trade-off

Safeguard	Benefit	Cost
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Safeguard	Benefit	Cost
RAG	Reduces hallucination	Retrieval latency
Human review	Improves safety	Requires human effort
Input validation	Prevents malformed inputs	Small processing overhead
Audit logging	Improves traceability	Storage overhead
Confidence checking	Identifies uncertainty	Additional computation

For a medical application, safety is prioritized over minimum latency; the design deliberately accepts a modest increase in response time in exchange for grounded, traceable, and reviewable output.

7. Broader Considerations

7.1 Reliability and Safety

- The system should never present an AI-generated explanation as a confirmed diagnosis.
- Display uncertainty wherever the model's confidence is not high.
- Show sources for every factual claim in the explanation.
- Identify extracted values distinctly from generated commentary.
- Flag high-risk cases for mandatory review.
- Recommend professional consultation for any concerning finding.
- Provide human review when necessary before treating an output as final.

7.2 Sustainability

- Small embedding models to minimize compute and energy usage.
- FAISS local retrieval instead of always-on cloud vector services.
- Top-3 retrieval to keep prompts and generation costs bounded.
- Smaller multimodal models where appropriate for the task.
- Avoiding unnecessary full-document processing when only a chunk is needed.

7.3 Accessibility

- Users can type questions for those who prefer text.
- Users can upload reports directly as images or PDFs.
- Users can speak questions for those with limited literacy or typing ability.
- Users can listen to generated explanations via optional text-to-speech.

7.4 Ethics and Professional Responsibility

- Protect patient privacy at every stage of processing and storage.
- Avoid unnecessary collection of personal information beyond what is required.
- Clearly communicate the system's limitations to every user.
- Avoid making definitive diagnoses under any circumstance.
- Support, rather than replace, healthcare professionals in the care pathway.

8. Conclusion

The proposed Gen AI Based Patient Explanation and Decision Support Module, delivered as a Generative AI-Enabled Multimodal Medical Report Explainer, combines RAG, vector databases, AI agents, vision-language models, speech processing, and responsible deployment practices into a single cohesive web application.

FAISS is selected as the vector database because it provides efficient and simple similarity search for a small medical knowledge corpus. A RAG pipeline improves the grounding of generated explanations by supplying relevant medical reference information to the LLM at generation time, rather than relying purely on the model's parametric memory.

For multimodal processing, text, medical report images, and speech are independently processed and combined using a late-fusion approach. This design provides flexibility and allows the system to continue operating even when one modality is unavailable, which is important for real-world robustness.

Responsible-AI safeguards such as source citations, confidence indicators, input validation, authentication, audit logging, privacy protection, prompt-injection handling, and human review collectively improve the reliability and trustworthiness of the system.

8.1 Requirements Satisfied

- RAG-based retrieval
- Vector database
- AI-agent reasoning
- Text processing
- Image processing
- Speech processing
- Multimodal fusion
- Streamlit interface
- Security controls
- Governance mechanisms
- Human-in-the-loop review
- Evaluation and validation

8.2 Limitations

- AI explanations may still contain errors despite grounding.
- Image quality can significantly affect extraction accuracy.
- Medical reference ranges vary across laboratories and populations.
- The system cannot replace professional diagnosis.
- Large-scale deployment would require stronger security and infrastructure.

8.3 Future Improvements

1. Support additional languages.
2. Add multilingual speech processing.
3. Improve OCR for scanned reports.
4. Use larger medical knowledge bases.
5. Improve medical entity extraction.
6. Add advanced hallucination detection.
7. Integrate hospital information systems securely.
8. Conduct formal medical-AI safety evaluation.

9. Student Reflection

Through this project, I learned how different generative AI technologies can be integrated into a single practical application. Previously, RAG, vector databases, multimodal AI, and responsible AI were studied separately. This project helped me understand how they interact as one complete system, and how a design decision in one layer — such as chunk size in retrieval — has downstream consequences for prompt length, latency, and answer quality.

I learned that retrieving information is not sufficient by itself. The retrieved information must be relevant, correctly passed to the LLM, and clearly presented to the user with provenance so that the explanation can be trusted and verified rather than taken on faith.

I also learned that multimodal systems require careful handling of different data types. Text, images, and speech have different processing requirements, failure modes, and quality characteristics, and missing or poor-quality inputs must be handled gracefully rather than causing the whole pipeline to fail.

Another important learning was responsible AI. In a medical application, accuracy and safety are more important than simply producing an answer quickly. Confidence indicators, source citations, privacy protection, and human review are therefore not optional extras but essential parts of the system architecture.

10. References

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6. FAISS Documentation, Facebook AI Similarity Search.
7. Sentence Transformers Documentation, Sentence-Transformers: Multilingual and Semantic Text Embeddings.
8. OpenAI Documentation, Generative AI and Multimodal Model Documentation.
9. Python Software Foundation, Python Documentation.
10. GitHub Documentation, Version Control and Repository Management.

Appendix A — Suggested Project Module Structure

Medical_Report_Explainer/

- app.py
- requirements.txt
- data/
 - medical_documents/
 - sample_reports/
- models/
 - embeddings.py
 - rag.py
 - vision.py
 - speech.py
 - agent.py
- database/
 - faiss_index/
- security/
 - validation.py
 - authentication.py
- governance/
 - safety.py
 - logging.py
 - model_card.md
- README.md

Main Streamlit Pages

1. Medical Report Upload
2. RAG Knowledge Retrieval
3. AI Agent
4. Image Report Analysis
5. Speech Input
6. Multimodal Explanation
7. Responsible AI & Security
8. Evaluation & Results